

Litesh Perumalla

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EDUCATION

University of North Texas
Master of Science, Data Science
• **GPA:** 3.9

Jan 2024 - Dec 2025
Denton, Texas

EMPLOYMENT EXPERIENCE

Discovery Park Library Services
Academic Assistant

Aug 2024 - Present
Denton, Texas

- Delivered precise and timely assistance in-person, by phone, and virtually to over 100 students and faculty weekly, enhancing accessibility to library services and resources.
- Collaborated with faculty to identify relevant academic materials, which contributed to more effective teaching outcomes and impactful research efforts.

crypt0nest.io

Jun 2025 - Aug 2025
Austin, Texas

Ai/ML Research & Product Intern (Team Lead)

- Developed a baseline machine learning pipeline for a Crypto Portfolio Predictive System using Python and time-series forecasting techniques, ensuring seamless data integration from various APIs.
- Engineered features from volatility indicators, applying backtesting and mean-variance optimization for robust strategy evaluation.
- Integrated real-time and historical crypto data by managing NOSQL database systems.
- Led the transition to the LLM and AI agent track by designing a multi-agent architecture that incorporated retrieval-augmented generation techniques for automated reasoning.
- Constructed and deployed an MCP server and client solution interfacing with LLMs, APIs, and databases, highlighting strong algorithm design and optimization skills.
- Conducted research on hybrid AI systems by combining machine learning predictions with LLM-driven agents, leveraging vector databases like ChromaDB and FAISS, while utilizing cloud technologies on Google Cloud Platform.
- Developed a Multi-Agent Architecture with more than 5+ agents a orchestrator and automated workflow and integrated ml model signals to AI Agents for Decision Making and Reasoning.

PROJECT EXPERIENCE

Smart Tutor AI – Personalized Teaching Support

Jan 2025 - Present

- Led the design and development of an AI-powered tutoring system that delivers course-specific responses using validated academic materials, enhancing learning Faithfulness and relevance by semantic chunking, Hybrid Retrievers.
- Built an efficient retrieval pipeline using metadata tagging and vector search to improve content discoverability and user interaction.
- Designed and implemented a multi-agent architecture using LangGraph to orchestrate modular agents responsible for student ID capture, query classification, course-specific routing, and answer clarification—ensuring context-aware, personalized educational support.
- Developed a Streamlit-based multi-page user interface with authentication (email, Google login), per-user chat history, feedback collection (thumbs up/down), and a customizable course homepage with announcements and professor links.
- Implemented real-time evaluation and feedback loops using LangSmith and LangFuse to monitor system outputs, improve model accuracy, and facilitate continuous fine-tuning based on user interactions.
- Tools used: RAG, Ollama, llama Index, Chroma db, Sentence-transformers, Vector Embeddings, BERT, Stream lit.

Skin Cancer Detection Using a Convolutional Neural Network

Nov 2024

- Implemented a CNN model to classify skin lesions from over 10,000 medical images, achieving 85% accuracy in early detection.
- Applied effective tuning and data augmentation techniques to enhance the model's generalization and robustness.
- Tools used: Python, Matplotlib, Sci-kit learn, NumPy, pandas, TensorFlow, CNN.

TECHNICAL SKILLS

- **Natural Language Processing:** Text Parsing, Sentiment Analysis, Transformers, BERT
- **Deep Learning:** Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), TensorFlow, PyTorch, Keras
- **Generative AI:** RAG (Retrieval-Augmented Generation), AI Agents, Vector Database, Prompt Engineering, Embeddings
- **Big Data And Cloud Technologies:** Amazon Web Services, Google Cloud Platform
- **LLM:** OpenAI, Ollama, llama Index, Lang Chain, Lang Graph, Fine Tuning
- **Database Management:** PostgreSQL, MongoDB, Chroma db, FAISS
- **Programming Languages:** Python, R, SQL, JavaScript, TypeScript
- **Version Control & DevOps:** Git, APIs
- **Certifications:** Machine Learning Specialization- Coursera
- **Core Competencies:** Algorithm Design, Enthusiasm For Learning